

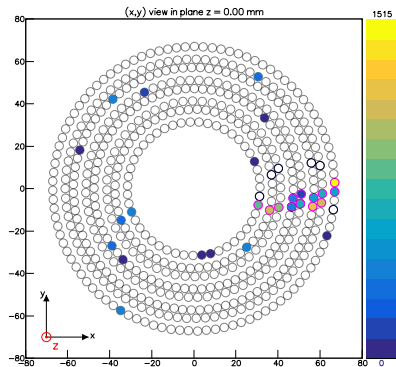
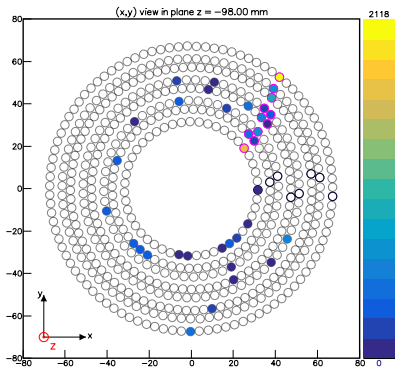
*ALERT meeting*

Reconstruction status  
bug report

Felix Touchte Codjo  
IJCLab

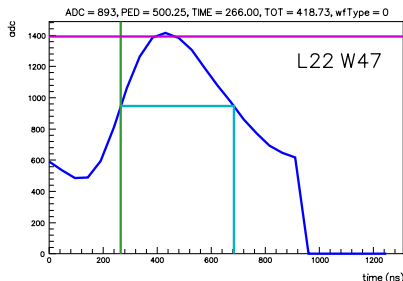
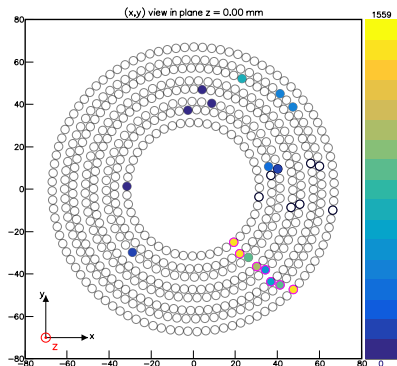
September 4, 2026

► Example of good tracks at 2.24 GeV, 50 nA

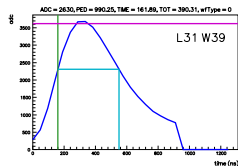
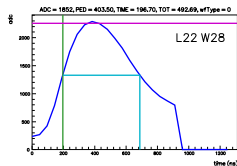
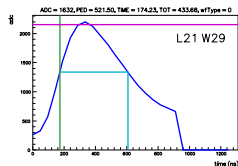
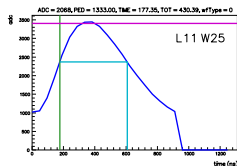
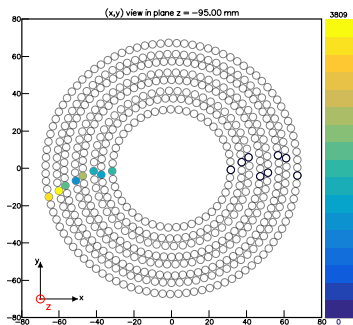


\*All hits highlighted in magenta belongs to a reconstructed track

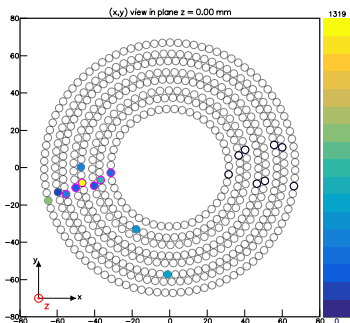
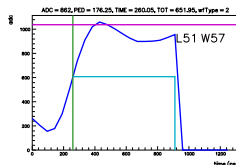
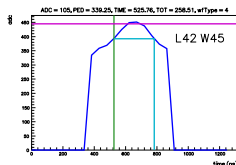
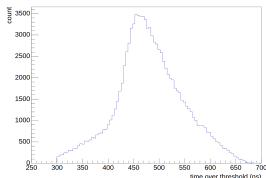
- Example of a good tracks with one missing hit
  - The waveform of this missing hit is plotted on the right
  - It pedestal does not pass the selection cut :  $120 < \text{PED} < 350$
  - Here the pedestal is 500.25 ADC



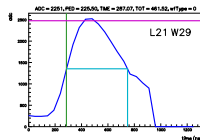
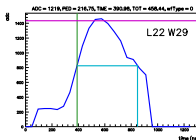
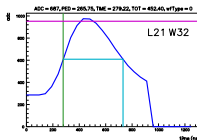
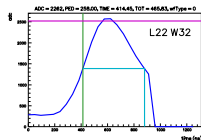
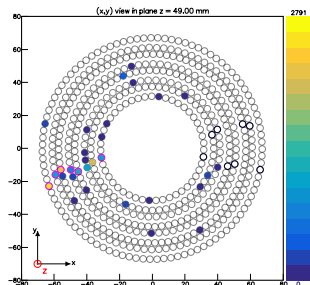
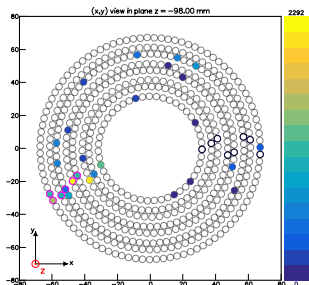
- ▶ Example of a potential good tracks not reconstructed due to the bad pedestal
  - Here all pedestals are above 350 ADC, so no hits were selected
  - Typical the signal in L11W25 is good, even if the baseline is around 1000 ADC
  - Reminder: during data acquisition, in mode TPC, we only kept, samples above threshold + few before and after (see appendix, last slide)
  - I think to remove the cut the pedestal



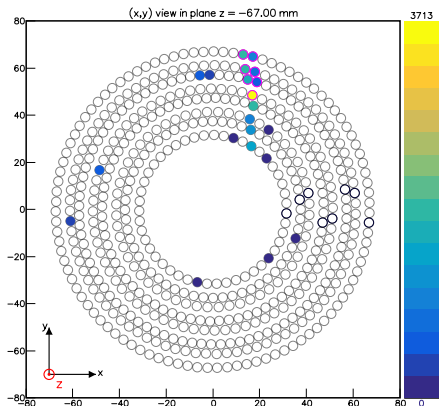
- ▶ I don't have much issue with the ToT, but I think to reduce the lower limit at 250 ns due to the apparent cut at 300 ns
  - In the example, the hits on layer 42 and 51 are missing
  - The hit on layer 42 has  $\text{ToT} < 300$  ns, so not selected
  - The hit on layer 51, has certainly been excluded in the second pass of the Kalman Filter by having a residual much bigger than 1.5 mm



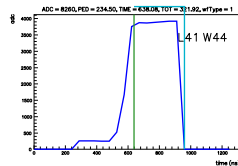
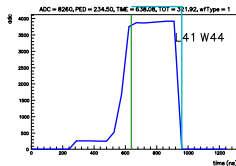
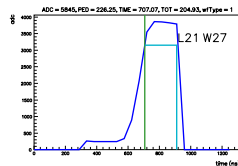
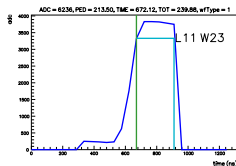
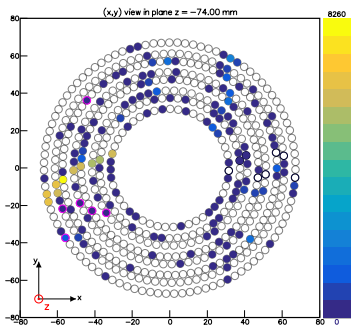
- Here, I looked at tracks with at least 6 hits, and I was surprised that we can have missing hits on a whole superlayer
  - The waveforms of these missing hits have no issue, there are all type 0
  - That happens a lot, they are mostly suppressed hits in the Kalman Filter
  - That suggests that we still need to have a better time and T2D calibration



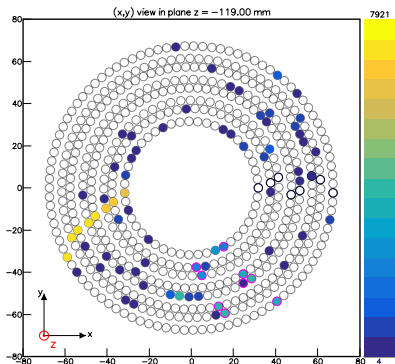
- ▶ Worst case scenario: not even a hit on the first 4 layers.
- ▶ In my next analysis, I will only consider tracks with at least 6 or 7 layers reached (before, I was just assuming that having more hits, meant reaching all layers)
- ▶ Until now, all these observations were made on run 22712, 2 GeV, 50 nA: things get worst at 10.6 GeV



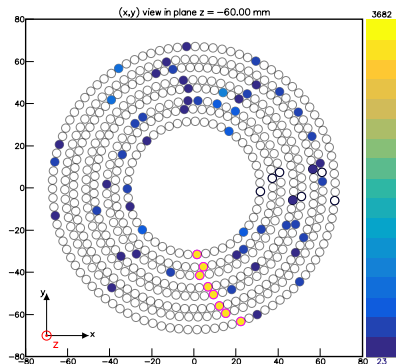
- ▶ Example on run 22172, 4He, 10.6 GeV, 325 nA
- ▶ Here I was expecting to reconstruct the high deposited energy
- ▶ Instead, I reconstructed an artifact (magenta circles)
  - Most of the saturated hits may seem late, I just think they need their own t0 calibration (what is not done yet)
  - Their ToT do not really make sense, and I should have removed the cut on the ToT for these signals



- ▶ Left, the reconstructed track is not an artifact, but we do not reconstruct the track with the high deposited energy (as before, excluded due to ToT lower than 300 ns)
  - It seems systematic to all data cooked post p0v11
- ▶ In p0v10\_3, we sometimes manage to catch the high deposited energy

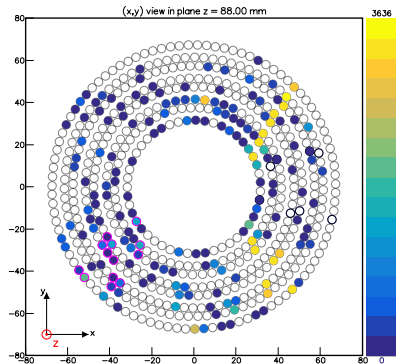
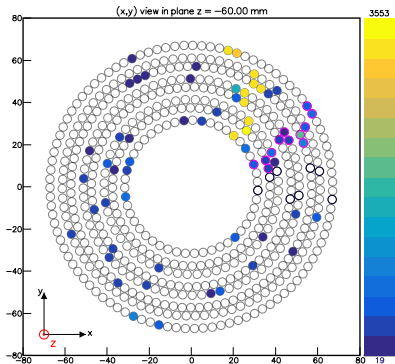


p0v12



p0v10\_3

- ▶ I have added some bias in p0v12, but even in p0v10\_3, we miss potential good tracks
  - Here, the ToT were in the good interval
  - A possible reason is that their time reconstruction was not good enough (again, the time should be better in p0v12, but we still have a ToT cut that makes no sense)



- ▶ The track reconstruction is well controlled at 2.24 GeV and low beam intensity
  - Track hits are easily selected
  - The issue encountered can be easily fixed by removing the pedestal and updating the ToT threshold
- ▶ The track reconstruction for 10.6 GeV run is difficult
  - The hit selection should be revisited
  - The t0 and T2D should be provided
  - We still reconstruct tracks, but some of them are artifacts
  - I plan to investigate those events by looking at elastic events off 4He

## Zero suppression

@CLAS12

- MVT – currently in firmware

- Fixed relationship between trigger and signal

- Some walk is possible

- Programmable **ZS window**

- Do not compare sample 1 to threshold : pedestal
- Compare samples 2-3, 2-3-4 or 2-3-4-5 to threshold  
→ Ex : samples 2 and 3

- Read all samples within the small **readout window**

- Ex: 9 samples

- TPC – to be implemented

- Signal can arrive anytime in the large readout window

- Continuously compare to threshold

- Keep samples above threshold + few before and after

